

Predictive Grid Stress Diagnostics via Non-Commutativity of Power Flow Jacobians

A Two-Channel Architecture for Cascade Detection and Localized Stress Identification

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Abstract—We derive a grid stress diagnostic based on the non-commutativity of the power flow Jacobian $\mathbf{J}$ and its time derivative $\dot{\mathbf{J}}$. The grid stress functional $\Lambda_G = \left\| [\mathbf{J}, \dot{\mathbf{J}}] \right\|_F$ measures sensitivity conflict—when the grid’s response to control actions evolves in incompatible directions.

While Λ_G converges to standard singularity metrics ($\lambda_{\min}$, L-index) under uniform stress, it demonstrates distinct advantages in three critical scenarios:

- 1) **Localized stress pockets:** When stress concentrates in a subset of buses, global metrics ($\lambda_{\min}$) show minimal response (−1.35%) while local Λ_G exhibits a 45× stronger response (+60%), preserving signal that global metrics average out.
- 2) **Topology discontinuities:** A shock-detection channel ($S = \|\Delta\mathbf{J}\|_F / \|\mathbf{J}\|_F$) provides +13.8s lead time before cascade collapse, while smooth trend monitoring fails completely.
- 3) **Oscillatory sensitivity:** Under dynamic loading (0.5Hz modulation), Λ_G responds with order-unity relative amplitude while $\lambda_{\min}$ is quasi-static ($< 10^{-5}$ relative change), on both meshed and radial topologies.

We recommend a two-channel alarm architecture combining trend monitoring for gradual degradation with shock detection for topology changes. Critically, we distinguish between stress localization (identifying the geometric source via Λ_G) and margin preservation (defending the binding constraint via $\min |V|$). Our intervention tests show that emergency response should target the weakest node, while preventive maintenance may benefit from targeting high- Λ_G regions.

Validation spans two fundamentally different topologies (IEEE 118-bus meshed transmission and IEEE 33-bus radial distribution) and includes a low-inertia configuration (40% renewable penetration) demonstrating that Λ_G is amplified 1.4–2.4× under reduced inertia. Λ_G is computationally cheaper than eigenvalue-based alternatives (3.1× faster on 118-bus, 4.4× on 300-bus) and achieves 7× lower false alarm rate under realistic stochastic load noise, with a

3-sample persistence filter eliminating all nuisance alarms.

Index Terms—Power system stability, Jacobian analysis, voltage collapse, cascade detection, localized stress, adaptive monitoring, non-commutativity

I. INTRODUCTION

A. The Problem

Modern power grids need diagnostics that can:

- 1) Provide early warning of impending instability
- 2) Localize stress to specific regions
- 3) Guide effective intervention

Traditional metrics (eigenvalue margin, voltage magnitude) excel at some of these tasks but fail at others.

B. The Key Insight: Diagnosis $\neq$ Control

Our validation revealed a fundamental distinction: the bus with the strongest stress *signal* is not necessarily the best target for corrective action. Λ_G identifies where stress originates; margin preservation requires targeting the binding constraint.

- **Preventive maintenance** → Target high- Λ_G regions (stress source)
- **Emergency intervention** → Target $\min |V|$ buses (binding constraint)

C. What Λ_G Does and Doesn’t Do

Table I summarizes Λ_G ’s validated performance across all test scenarios, including cross-topology and low-inertia configurations. The remainder of this paper substantiates each row.

TABLE I
 Λ_G PERFORMANCE SUMMARY

Capability	Performance	Evidence
Uniform warning	$= \lambda_{\min}$	Tied at 151 s
Cascade warning	+13.8 s advantage	Shock channel
Localized stress	+60% vs -1.35%	$45\times$ sensitivity
Oscillatory	Order-1 vs quasi-static	Both topologies
Emergency	$< \min V $	0 vs +29.9 MW

D. Contributions

- 1) **Two-channel alarm architecture** (Trend + Shock) for comprehensive coverage
- 2) **Localized stress detection** where global metrics fail
- 3) **Honest empirical evaluation** distinguishing diagnosis from control
- 4) **Practical recommendations** for when to use which metric
- 5) **Oscillatory sensitivity:** Under dynamic loading (0.5 Hz modulation), Λ_G responds with order-unity relative amplitude while $\lambda_{\min}$ remains quasi-static ($< 10^{-5}$ relative change), on both meshed and radial topologies (Section III-C).
- 6) **Operational advantages:** Λ_G is $3.1\text{--}4.4\times$ faster than $\lambda_{\min}$ (118–300-bus), achieves $7\times$ lower sample-level FAR, and a 3-sample persistence filter eliminates all nuisance alarms (Section VII-C).

II. MATHEMATICAL FRAMEWORK

A. The Grid Stress Functional

$$\Lambda_G(t) = \left\| [\mathbf{J}(t), \dot{\mathbf{J}}(t)] \right\|_F \quad (1)$$

The commutator measures **sensitivity conflict**—when control sensitivities evolve in incompatible directions.

Scaling. The Jacobian $\mathbf{J}$ from Newton–Raphson power flow has rows for P/Q (MW/MVAR) and columns for θ/V (rad/pu), so Λ_G inherits the per-unit base. We do not apply block scaling; all results use standard per-unit conventions. Λ_G is *not* invariant to per-unit base choice—thresholds must be recalibrated if the base changes. The same is true of $\lambda_{\min}$ and S . The sensitivity *ratios* reported (e.g., $45\times$, $3.1\times$ faster) are dimensionless and base-independent.

Computation of $\dot{\mathbf{J}}$. The time derivative $\dot{\mathbf{J}}$ is computed via element-wise Savitzky–Golay differentiation (polynomial order 2, window 0.4 s) applied to a time series of Jacobians extracted from successive AC power flow solutions. This requires **PMU-class sampling (10–60 Hz)**; SCADA refresh rates (2–4 s, i.e., 0.25–0.5 Hz sampling) provide insufficient temporal resolution because a 0.4 s SG window contains fewer than two samples at those rates, collapsing to noisy finite difference. Our validation uses 10 Hz (development) and 30 Hz (confirmation), both yielding identical lead times and metric rankings (Section VI-A), confirming robustness across the PMU-rate band.

For substations lacking PMU infrastructure, the shock channel ($S = \|\Delta\mathbf{J}\|_F / \|\mathbf{J}\|_F$, Eq. 3) remains viable at SCADA rates since it requires only two consecutive Jacobian snapshots rather than a smooth derivative. This makes the two-channel architecture deployable in mixed PMU/SCADA environments, with the trend channel active only at PMU-equipped substations.

B. Bus-Level Decomposition

Let $\mathbf{C} = [\mathbf{J}, \dot{\mathbf{J}}]$ denote the commutator matrix ($n \times n$, where $n = 2N_{\text{bus}}$ for a P - Q formulation). Define $\mathcal{I}_i = \{2i-1, 2i\}$ as the row/column indices associated with bus i . Then

$$\Lambda_G^{(i)} = \sqrt{\sum_{j \in \mathcal{I}_i} \sum_k C_{jk}^2 + \sum_j \sum_{k \in \mathcal{I}_i} C_{jk}^2} \quad (2)$$

collects all commutator energy in the rows and columns belonging to bus i . Physically, $\Lambda_G^{(i)}$ measures how much the sensitivity conflict at bus i contributes to the global Λ_G ; buses with large $\Lambda_G^{(i)}$ are where control-action responses are evolving most incompatibly.

C. The Shock Metric

$$S(t) = \frac{\|\mathbf{J}(t) - \mathbf{J}(t - \Delta t)\|_F}{\|\mathbf{J}(t - \Delta t)\|_F + \varepsilon} \quad (3)$$

where $\varepsilon = 10^{-6}$ prevents division by zero. Since $\|\mathbf{J}\|_F > 500$ for both test systems, $\varepsilon / \|\mathbf{J}\|_F < 10^{-9}$ —entirely negligible. The shock detection threshold is calibrated separately (Appendix A). S detects topology discontinuities (line trips, generator outages) that trend metrics miss.

III. LOCALIZED STRESS DETECTION

A. The Problem with Global Metrics

Global metrics like $\lambda_{\min}$ compute an aggregate over the entire system. When stress concentrates in a small region, the local signal is averaged out by the quiescent majority.

B. Quantitative Comparison

TABLE II
RESPONSE TO LOCALIZED STRESS (IEEE 118-BUS)

Metric	Response	Interpretation
$\lambda_{\min}$	−1.35%	Within normal variability
L_{index}	−0.8%	Within normal variability
Λ_G (local)	+60%	45× stronger response

The baseline distributions of all metrics are non-Gaussian (Shapiro–Wilk $p < 10^{-30}$ for all), precluding sigma-level reporting. Instead, we characterize Λ_G ’s advantage as a **sensitivity ratio**: the local $\Lambda_G^{(i)}$ response to weak-pocket stress is 45× larger than the $\lambda_{\min}$ response (60%/1.35%), demonstrating that Λ_G preserves localized signal that global metrics average out.

For cascade events (Section VI-B), bus-level $\Lambda_G^{(i)}$ deviations are genuinely extreme: z -scores of 55–161 relative to the normal baseline after topology changes, exceeding the 99.99th empirical percentile.

C. Oscillatory Sensitivity

In dynamic regimes (0.5 Hz load modulation, 10% amplitude, 20 Hz sampling), Λ_G acts as a high-sensitivity sensor for Jacobian frame rotation ($\mathbf{J}$). Table III reports the peak-to-peak amplitude normalized by the mean for each metric.

TABLE III
OSCILLATORY SENSITIVITY (0.5 HZ, CROSS-TOPOLOGY).
RELATIVE AMPLITUDE = PEAK-TO-PEAK / MEAN. $\lambda_{\min}$ IS
QUASI-STATIC, MAKING THE RATIO DENOMINATOR-DEPENDENT.

System	Topology	Λ_G Rel.	$\lambda_{\min}$ Rel.	Ratio
118-bus	Meshed	1.476	9.2×10^{-6}	161,000
33-bus	Radial	1.558	3.1×10^{-6}	508,000

The key observation is not the ratio per se—which is large because $\lambda_{\min}$ is essentially quasi-static under oscillatory loading—but rather that Λ_G

responds with order-unity relative amplitude (1.48–1.56) while $\lambda_{\min}$ shows no measurable oscillatory response ($< 10^{-5}$). This qualitative distinction holds on both meshed and radial topologies, confirming that Λ_G captures the **volatility** of the system state in a way that eigenvalue-based metrics cannot. Unlike the localized stress advantage (which is meshed-specific, see Section VI-D), the oscillatory sensitivity advantage is topology-independent.

IV. TWO-CHANNEL ALARM ARCHITECTURE

A. Why Two Channels?

TABLE IV
CHANNEL COMPLEMENTARITY

Failure Mode	Trend Channel	Shock Channel
Gradual degradation	✓ Detects	× Misses
Topology discontinuity	× Late (−10 s)	✓ Early (+13.8 s)

Neither channel alone is sufficient.

B. Architecture

The two-channel alarm system operates as follows:

- **Trend channel:** Metric Λ_G (or $\lambda_{\min}$); Logic: $> \Theta$ for k consecutive samples; Detects gradual decay.
- **Shock channel:** Metric $S = \|\Delta \mathbf{J}\|_F / \|\mathbf{J}\|_F$; Logic: $> \Theta$ (instant); Detects topology change.
- **Global alarm:** OR of both channels.

C. Validated Performance

TABLE V
ALARM CHANNEL PERFORMANCE (CASCADE TEST)

Channel	Lead Time
Shock	+13.8 s ✓
Trend alone	−10.0 s ×

Shock threshold (10^{-5}) calibrated to 99.5th percentile of benign switching events.

V. DIAGNOSIS VS. CONTROL: THE STRATEGIC DISTINCTION

A. The Core Finding

Test 3.1 revealed that **targeting the stress source** $\neq$ **saving the grid**: injecting 50 MVAR at the $\text{Max-}\Lambda_G$ bus (Bus 13) yields 0.00 MW margin improvement, while targeting $\min |V|$ (Bus 37) yields +29.86 MW. For uniform stress, the collapse bottleneck is the binding constraint (weakest node), not the strongest source.

B. The Hybrid Score: Uniform and Localized Stress

We implemented and validated a hybrid scoring function:

$$\text{score}(i) = \frac{\kappa_i}{(V_i - V_{\text{crit}})^2} \quad (4)$$

where $\kappa_i = \|dV/dQ\|_2$ (controllability).

We validate the hybrid score under both uniform and non-uniform stress:

Uniform stress (180% all buses): The hybrid score converges to the $\min |V|$ strategy (selecting Bus 37), confirming that it safely prioritizes the binding constraint while incorporating controllability.

Localized stress (150% global, 180–210% on 15-bus weak area): $\text{Max-}\Lambda_G$ selects Bus 45 (outside stressed region, +0 MW at all stress levels), $\min |V|$ selects Bus 74/117 (+34 to +108 MW), and the hybrid converges to $\min |V|$ only at critical stress. The hybrid score is therefore not recommended for operational deployment; $\min |V|$ alone is sufficient for emergency targeting. Λ_G 's role remains diagnostic: identifying WHERE stress originates for preventive maintenance, not WHERE to inject.

VI. VALIDATION SUMMARY

A. Test 1: Uniform Load Ramp (IEEE 118-Bus)

TABLE VI
UNIFORM LOAD RAMP LEAD TIMES

Metric	Lead Time	Relative to Λ_G
Λ_G	151.2 s	baseline
$\lambda_{\min}$	151.2 s	equal
L_{index}	151.2 s	equal
$V_{\min}$	151.2 s	equal

Conclusion: Under uniform stress, Λ_G converges to traditional metrics.

B. Test 2: Cascade Reconstruction (2003 Northeast Blackout)

We reconstruct a simplified analog of the August 14, 2003 Northeast blackout cascade [7] on the IEEE 118-bus system. The original event involved generator trips (Eastlake Unit 5), transmission line losses (Harding–Chamberlin, Hanna–Juniper), and cascading relay operations over approximately 90 minutes. Our reconstruction maps this sequence to the 118-bus topology:

TABLE VII
2003 BLACKOUT MAPPING

Phase	2003 Event	118-Bus Analog
0	Normal under high load	140–180% ramp (0–60 s)
1	Eastlake Unit 5 trip	Gen trip at Bus 10 (60 s)
2	Harding–Chamberlin loss	Line trip 30–38 (360 s)
3	Hanna–Juniper loss	Line trip 24–70 (460 s)

While this is a stylized reconstruction (compressed timescale, different network topology), it preserves the essential failure mode: sequential topology changes creating stress propagation faster than operator awareness. Lead time results (Table V) confirm that the shock channel is essential for topology discontinuities.

C. Tests 3–4: Localized Stress and Intervention Targeting

Local $\Lambda_G^{(i)}$ increases +60% under weak-pocket stress while $\lambda_{\min}$ changes only -1.35% ($45\times$ sensitivity advantage). However, targeting the $\text{Max-}\Lambda_G$ bus for 50 MVAR injection yields 0.00 MW margin improvement versus +29.86 MW for $\min |V|$. Λ_G detects localized stress that global metrics miss, but emergency intervention should target the binding constraint (Section V).

D. Cross-Topology Validation: IEEE 33-Bus Radial System

To test generalizability, we repeat the uniform load ramp and localized stress tests on the IEEE 33-bus radial distribution system [8], which differs fundamentally from the 118-bus transmission network: radial topology (no loops), 33 buses, no synchronous generators (single external grid), and distribution-level voltages.

Uniform load ramp: All three metrics (Λ_G , $\lambda_{\min}$, $V_{\min}$) alarm simultaneously at $t = 60.9$ s

with 152.3 s lead time before collapse at 278% loading—consistent with the 118-bus finding that Λ_G converges to $\lambda_{\min}$ under uniform stress.

Localized stress (feeder-end loading): Unlike the meshed 118-bus results, $\lambda_{\min}$ changes -9.0% (visible signal) while local Λ_G increases only $+1.3\%$. On radial networks, there is no alternative path to absorb local stress, so global eigenvalue metrics detect localized problems more readily. This confirms that Λ_G 's localized stress advantage (Section III) is specific to **meshed topologies** where redundant paths average out local signals.

TABLE VIII
TOPOLOGY-DEPENDENT LOCALIZED STRESS RESPONSE

System	Topology	Λ_G vs $\lambda_{\min}$ (localized)
IEEE 118-bus	Meshed	$\Lambda_G \gg \lambda_{\min}$ (+60% vs -1.35%)
IEEE 33-bus	Radial	$\Lambda_G \ll \lambda_{\min}$ (+1.3% vs -9.0%)

This topology-dependent behavior is expected: Λ_G measures sensitivity *conflict* between coupled control paths, which is inherently a meshed-network phenomenon.

E. Spatial Stress Pattern During Cascade

Tracking bus-level $\Lambda_G^{(i)}$ during the cascade reconstruction reveals the spatial distribution of sensitivity conflict after each topology change.

TABLE IX
SPATIAL STRESS CONCENTRATION (IEEE 118-BUS CASCADE)

Phase	Event	Top 5 Stressed Buses	Peak z
1	Gen trip (Bus 10)	77, 13, 45, 26, 80	3.9
2	Line trip (30–38)	45, 26, 13, 77, 80	65
3	Line trip (24–70)	45, 26, 13, 77, 36	161

Stress concentrates at network hubs (buses 45, 26, 13, 77—major interconnection points), not at event locations: Bus 10 (trip site) never appears in the top 5. Peak z -scores escalate nonlinearly from 3.9 to 65 to 161 across three events ($40\times$ amplification), reproducing the cascade amplification mechanism of the 2003 blackout. Since electrical propagation is near-instantaneous, the diagnostic value of bus-level $\Lambda_G^{(i)}$ is identifying *which* buses are most vulnerable to further cascade.

F. N-k Cascade Screening via Bus-Level Rankings

To test whether $\Lambda_G^{(i)}$ rankings after the first contingency predict subsequent cascade vulnerability, we record bus-level rankings through a four-phase cascade: Phase 0 (160% stressed baseline) $\rightarrow$ Phase 1 (generator trip) $\rightarrow$ Phase 2 (line trip) $\rightarrow$ Phase 3 (second line trip).

TABLE X
SPEARMAN RANK CORRELATION OF BUS-LEVEL $\Lambda_G^{(i)}$

Comparison	Spearman ρ	Top-5 Overlap
Phase 0 $\rightarrow$ 1 (gen trip)	0.714	3/5
Phase 1 $\rightarrow$ 2 (line trip)	1.000	5/5
Phase 1 $\rightarrow$ 3 (two line trips)	0.993	5/5

The first contingency (generator trip) genuinely reshuffles the vulnerability landscape ($\rho = 0.71$ from baseline), but subsequent line trips preserve the post-contingency ranking with $\rho > 0.99$ and perfect top-5 overlap. This means $\Lambda_G^{(i)}$ computed after the first event is essentially a **perfect predictor** of which buses will be hit by the next event—a property directly applicable to real-time N-k screening.

G. Low-Inertia Validation: High Renewable Penetration

To assess Λ_G performance under conditions representative of future grids, we modify the IEEE 118-bus system by replacing 40% of synchronous generators with static generators (modeling inverter-based resources without inertia) and reducing reactive power limits on remaining generators to 70% of nominal.

Voltage stability margin. The low-inertia configuration collapses at 170% uniform loading versus 182% for full inertia, a 12 percentage point reduction consistent with reduced reactive support. Λ_G at matched loading levels is amplified $1.07\text{--}1.41\times$, providing earlier warning precisely when stability margins are thinner.

Localized stress. Under weak-pocket stress, Λ_G amplification ranges from $1.73\times$ (120% local) to $2.37\times$ (200% local), with the low-inertia system identifying different critical buses, reflecting changed power flow patterns from generator displacement. Both configurations maintain $\Lambda_G \gg \lambda_{\min}$ for oscillatory detection ($10\times$ full, $7\times$ low inertia). These ratios are smaller than those in

TABLE XI
LOAD RAMP: FULL VS. LOW INERTIA

Load Level	Full Inertia Λ_G	Low Inertia Λ_G	Amplification
120%	1626	1737	1.07×
140%	1977	2246	1.14×
160%	2838	4011	1.41×
Collapse	182%	170%	−12 pp

Table III (161,000×) because the reduced-inertia system operates closer to its stability boundary ($\lambda_{\min}$ mean = 0.158), making $\lambda_{\min}$ more responsive to perturbations; Λ_G 's relative amplitude is identical (≈ 1.48) in both configurations, confirming that the sensitivity ratio is operating-point-dependent while Λ_G 's oscillatory response is not. These results confirm that Λ_G is **amplified** under high renewable penetration.

VII. PRACTICAL RECOMMENDATIONS

A. For Grid Operators

TABLE XII
METRIC SELECTION GUIDE

Scenario	Recommended Metric
Early warning (gradual)	Λ_G or $\lambda_{\min}$ (equivalent)
Early warning (cascade)	Shock channel (S)
Localized stress	Local $\Lambda_G^{(i)}$
Emergency injection	$\min V $ (binding constraint)
Preventive maintenance	$\max \Lambda_G$ (stress source)

TABLE XIII
DATA REQUIREMENTS BY DEPLOYMENT

Scenario	Metric	Data Rate
PMU-equipped	Λ_G trend + Shock	≥ 10 Hz
SCADA-only	Shock channel only	≥ 0.25 Hz

B. Operator Display

The recommended operator interface presents both diagnostic (source identification via Λ_G) and control (binding constraint via $\min |V|$) information simultaneously, with distinct recommended actions for immediate emergency response (inject at $\min |V|$ bus) and follow-up preventive action (investigate $\max \Lambda_G$ region).

C. Computational Cost and Robustness

Table XIV reports per-timestep computation cost (100-step average, 10 Hz).

TABLE XIV
COMPUTATIONAL COST PER TIMESTEP (MS, CONVERGED STEPS ONLY)

Component	33-bus	118-bus	300-bus*	33→300
Power flow	9.7	15.9	18.8	1.9×
Eigendecomp	0.7	36.2	201	271 ×
Commutator	0.1	1.1	31	310×
Total Λ_G	9.8	17.0	49.8	5.1×
Total $\lambda_{\min}$	10.4	52.1	219.8	21.1×
Speedup	1.1×	3.1×	4.4 ×	—

*33% PF convergence rate; timings from converged steps only.

Λ_G 's speedup over $\lambda_{\min}$ grows with system size: 1.1× (33-bus), 3.1× (118-bus), **4.4**× (300-bus). At 300-bus, eigendecomposition (201 ms) dominates the $\lambda_{\min}$ pipeline, while the commutator (31 ms) remains comparable to power flow. The commutator cost matches expected cubic scaling in Jacobian dimension ($530/181 \approx 2.9$; $2.9^3 \approx 25$; measured 28×), consistent with dense matrix multiplication. Both metrics remain real-time-capable at 10 Hz through 300 buses.

SG window robustness. Sweeping the Savitzky–Golay window from 0.2 s to 1.0 s at 10 Hz produces lead time variation < 0.1 s and false-positive count change ≤ 1 sample (out of 488), confirming that the 0.4 s default is not a sensitive parameter.

Jacobian estimation error. PMU channel dropout (randomly zeroing 5–20% of Jacobian entries) degrades Λ_G magnitude by only 6–20%, with bus-level ranking stability $\rho > 0.97$ (Spearman), demonstrating graceful degradation under partial observability. Element-wise Gaussian noise is more adversarial: 0.1% noise produces 5× relative error in Λ_G under raw finite difference because the noise dominates the small inter-step ΔJ . However, bus-level rankings remain correlated ($\rho \approx 0.6$), meaning $\Lambda_G^{(i)}$ still identifies the correct stressed buses even when its magnitude is unreliable—motivating the SG smoothing filter already in our pipeline. The FAR Monte Carlo confirms that realistic load noise propagated through power flow (a more physically faithful noise model) yields sub-1% FAR.

False alarm rate. Under realistic autocorrelated load noise (3% σ , 30 s time constant, 50 Monte Carlo trials at 10 Hz), Λ_G achieves 0.7% sample-

level FAR and shock 0.7% at 99.5th-percentile thresholds. $\lambda_{\min}$ exhibits 4.2%— $7\times$ higher. Sample-level FAR does not translate directly to operational nuisance rate at high sampling rates; we define **event-level** alarms using a k -consecutive exceedance rule. With $k = 3$ (0.3 s persistence at 10 Hz), nuisance alarms drop to **zero events/hr** across all 50 trials for both Λ_G and shock, adding only 0.2 s detection latency—a negligible cost for eliminating all false alarms.

VIII. DISCUSSION

A. When Λ_G Provides Value

Table XV summarizes where Λ_G adds value beyond existing metrics and where it does not. The normalized metric Ξ_G proved non-discriminative and was excluded.

TABLE XV
 Λ_G ADVANTAGE BY SCENARIO

Scenario	Λ_G Value	Mechanism
Localized (meshed)	High	Global metrics average out
Topology shock	High	Shock channel catches
Low inertia	High	1.4–2.4 $\times$ amplified
Uniform drift	None	Converges to $\lambda_{\min}$
Radial distrib.	Low	Single path sufficient
Emergency inject	None	$\min V $ dominates

B. Algebraic Structure

The commutator $[\mathbf{J}, \dot{\mathbf{J}}]$ equals twice the antisymmetric (wedge) product $\mathbf{J} \wedge \dot{\mathbf{J}}$, so $\Lambda_G = 2 \|\mathbf{J} \wedge \dot{\mathbf{J}}\|_F$ measures the bivector magnitude—the oriented area swept by the Jacobian frame as the operating point evolves. This connects to the Schur decomposition: the internal term $[\mathbf{T}, \dot{\mathbf{T}}]$ captures eigenvalue dynamics while the frame rotation term $[\mathbf{T}, [\boldsymbol{\Omega}, \mathbf{T}]]$ captures eigenvector precession. Future work may exploit this grading for spectral-flow decomposition theorems [10].

IX. CONCLUSION

We have developed and validated a grid stress diagnostic based on the non-commutativity of the power flow Jacobian. Across two topologies (IEEE

118-bus meshed, 33-bus radial) and two inertia configurations (full synchronous, 40% renewable), Λ_G provides distinct advantages where existing metrics fall short: localized stress detection (45 $\times$ sensitivity ratio on meshed networks), topology-shock early warning (+13.8 s lead time via shock channel), and oscillatory sensitivity (order-unity response where $\lambda_{\min}$ is quasi-static). Under uniform stress it converges to $\lambda_{\min}$, confirming consistency. Under low inertia, Λ_G is amplified 1.4–2.4 $\times$, making it increasingly relevant as synchronous generation is displaced by renewables.

Importantly, diagnosis is not control: emergency intervention should target $\min |V|$ (the binding constraint), while Λ_G identifies stress origins for preventive maintenance. The localized stress advantage is specific to meshed topologies; on radial networks, $\lambda_{\min}$ suffices. These limitations, reported alongside the strengths, define the operational scope of the two-channel architecture.

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APPENDIX A

THRESHOLD CALIBRATION (IEEE 118-BUS)

Two calibration contexts appear in this paper; they are complementary, not contradictory.

Ramp-test thresholds (Sections III–VI): set as the 95th percentile of each metric during the first 30% of a uniform load ramp (normal-operation baseline), yielding 5% sample-level FAR during the

calibration window. These are intentionally loose to capture the earliest possible alarm.

TABLE XVI
RAMP-TEST THRESHOLDS (95TH PERCENTILE)

Metric	Threshold	Cal. FAR
Λ_G	27.85	5%
S (shock)	10^{-5}	5%
$\lambda_{\min}$	0.160	5%

Noise-test thresholds (Section VII-C): set as the 99.5th percentile of each metric under 50 Monte Carlo trials of autocorrelated load noise (3% σ , 30 s time constant). These are tighter, yielding sub-1% sample-level FAR for Λ_G (0.64%) and S (0.50%), but 4.2% for $\lambda_{\min}$ due to its heavier-tailed noise distribution. Operational deployment should use the noise-calibrated thresholds; ramp-test thresholds are for experimental comparison only.

APPENDIX B INTERVENTION DETAILED RESULTS

Strategy	Target	Margin
Max Λ_G	Bus 13	0.00 MW
Max div	Bus 13	0.00 MW
Min V	Bus 37	+29.86 MW

APPENDIX C GLOSSARY

Term	Definition
Sensitivity conflict	Control sensitivities evolving in incompatible directions
Shock metric (S)	$\ \Delta \mathbf{J}\ _F / \ \mathbf{J}\ _F$
Binding constraint	Weakest element limiting system margin
Two-channel	Trend + Shock monitoring combined
Sensitivity ratio	Relative response of Λ_G vs $\lambda_{\min}$